

Generative AI and creative self-efficacy: A systematic review

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ABSTRACT

Creative self-efficacy shapes motivation, persistence, and engagement in creative work. As generative artificial intelligence systems become increasingly integrated into educational, professional, and artistic contexts, questions have emerged regarding how these technologies influence individuals' creative self-beliefs. This systematic review synthesizes research published between 2021 and 2025 to examine how generative AI is associated with creative self-efficacy, creative confidence, and related constructs during a period of widespread adoption. Following PRISMA guidelines, we analyzed 42 peer-reviewed studies across educational, professional, and creative domains. Fourteen studies directly examined relationships between the use of generative AI and creative self-beliefs, while additional studies provided contextual insight into how creative self-efficacy was conceptualized and measured during the AI integration period. Findings across the reviewed literature were heterogeneous, varying across domains and study designs.

Across studies, recurring patterns indicate that perceived control, attribution clarity, and opportunities for human agency are frequently associated with differences in reported creative confidence and self-efficacy. When generative AI was positioned as a supportive tool within transparent co-creation processes, studies more often reported neutral or positive associations with creative self-beliefs. When AI outputs were perceived as displacing human authorship or obscuring individual contributions, studies more commonly reported uncertainty or reduced confidence. Bandura's four sources of self-efficacy remain relevant in AI-mediated contexts but appear to operate differently in human–AI collaboration. Overall, the review suggests that generative AI does not uniformly enhance or diminish creative self-efficacy; its psychological implications depend on contextual, pedagogical, and design-related factors.

Creative self-efficacy and creative confidence are foundational psychological constructs essential for understanding human engagement in creative processes across domains. Rooted in social cognitive theory and mindset theories, these constructs shape motivation, persistence, and performance, with significant implications for well-being and adaptive functioning (Bandura, 1977, 2001; Dweck, 2006).

The rapid emergence of generative artificial intelligence tools represents a profound shift in creativity research and education. Unlike prior tools that primarily enhanced human creativity, generative AI autonomously produces creative outputs that rival or surpass human performance in specific domains (Marrone et al., 2024).

For the purposes of this review, generative artificial intelligence (generative AI) refers to AI systems designed to generate novel

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content, including text, images, music, and other artifacts, in response to human prompts, consistent with established usage in the computer science and HCI literature (Cao et al., 2023). Throughout this review, "AI" refers specifically to generative AI unless otherwise noted.

Systems such as ChatGPT, DALL-E, and Midjourney can generate increasingly sophisticated text, images, music, and code, prompting renewed scholarly debate about the boundaries of human creativity and the status of creative authorship (Marrone et al., 2024). These developments have raised questions about how individuals may understand their creative role when working alongside generative systems. For educators, this raises critical questions about how to foster creative thinking and confidence when students have access to tools that can independently produce creative work.

In educational settings, teacher feedback that emphasizes experimentation and iterative improvement has traditionally enhanced students' creative self-efficacy (Beghetto, 2010). Pedagogical approaches such as experiential learning can offer mastery experiences that foster confidence and academic engagement (Beghetto, 2010; Creely et al., 2021; Payne & Whitworth, 2022; Wingard et al., 2022). However, as AI becomes increasingly integrated in classrooms and creative workplaces, understanding how these technologies affect creative self-beliefs becomes essential for designing effective learning environments.

Despite considerable advances in understanding creative confidence and self-efficacy, their relationship with AI has been under-explored. This review addresses this gap by examining research from 2021 to 2025, a pivotal period marked by explosive growth in AI capabilities and adoption. Understanding how AI influences creative self-beliefs is essential for educators, curriculum designers, and anyone concerned with fostering creative thinking in contemporary learning environments, where human-AI collaboration is becoming the norm rather than the exception.

1. Background

1.1. Creative self-efficacy and creative confidence

In this background section, we first define and distinguish creative self-efficacy (CSE), creative confidence (CC), and creative agency (CA), the primary constructs examined in this review, and explain how they are hierarchically interrelated. We then review Bandura's four sources of self-efficacy as the principal theoretical lens through which we interpret generative AI's influence on creative self-beliefs throughout this review. Finally, we examine how generative AI systems interact with these constructs and consider the environmental and contextual factors that moderate their development. Although all three constructs are introduced in the Background, creative self-efficacy receives the most emphasis throughout the review because it is the construct most directly and consistently operationalized and measured in the reviewed empirical literature. Where evidence permits, we draw theoretically grounded inferences for the broader constructs of creative confidence and creative agency, making the inferential step explicit in each case.

Creative confidence refers to a general belief in the ability to generate novel and valuable ideas and a willingness to pursue creative endeavors. Creative self-efficacy refers to domain- and task-specific beliefs about one's capacity to succeed in creative endeavors (Beghetto, 2006; Tierney & Farmer, 2002). While related, these constructs differ in important ways. Creative self-efficacy is more malleable and situational, while creative confidence represents a more enduring characteristic that reflects broader beliefs about one's creative abilities across contexts.

These distinctions have methodological implications. Measures tailored to specific tasks or domains provide more predictive validity for creative performance in those contexts than general assessments (Beghetto et al., 2021). This specificity matters particularly in mediated contexts, where individuals may feel confident using AI for certain creative tasks but uncertain about others.

These constructs operate in a dynamic feedback loop. Specific task successes build broader creative confidence, while general confidence motivates engagement in new creative challenges. This means that AI experiences affecting task-specific self-efficacy may ultimately influence broader creative identity. In AI-mediated contexts, these constructs are shaped by how people interpret their own contributions relative to system-generated outputs. Table 1 summarizes these distinctions and their relevance to AI contexts.

Table 1
Relationship between creative-self-efficacy and creative confidence.

Dimension	Creative Self-Efficacy	Creative Confidence	Relationship/Interaction
Definition	Domain-specific belief in ability to create	Generalized belief in creative ability	Self-efficacy experiences inform broader confidence
Scope	Context- and task-specific	Cross-domain, relatively stable trait	Specific successes build general confidence; general confidence influences task approach
Stability	Malleable and situational	More enduring characteristic	Changes in self-efficacy can shift confidence over time; confidence provides foundation for self-efficacy
Measurement	Task-specific scales	General creative confidence scales	Both needed for comprehensive assessment; self-efficacy predicts performance, confidence predicts engagement
Theoretical Basis	Bandura's self-efficacy theory	Broader creative identity and mindset	Self-efficacy mechanisms (mastery, vicarious learning) feed into confidence; confidence shapes willingness to attempt new creative tasks
AI Context Relevance	Directly affected by specific human-AI interactions	Influenced by accumulated AI experiences and identity negotiations	AI success in specific domains builds efficacy; repeated efficacy experiences reshape overall creative confidence

1.2. Bandura's four sources of self-efficacy

Bandura's (1977, 2001) social cognitive theory identifies four sources through which self-efficacy beliefs develop. Mastery experiences (direct experiences of success or failure in performing tasks) are the most influential source. Successful task completion strengthens self-efficacy, while repeated failures weaken it. Vicarious learning occurs through observing others successfully perform tasks, particularly those perceived as similar to oneself. Social persuasion encompasses verbal encouragement, feedback, and evaluations from others. Affective and physiological states (emotions, moods, and physical sensations experienced during task performance) also inform efficacy judgments. These four sources interact dynamically, with mastery experiences typically having the strongest impact.

In creative contexts, successfully completing creative challenges builds creative self-efficacy. Seeing peers succeed at creative tasks can boost confidence through social comparison. Teacher and peer feedback that emphasizes growth and specific improvements strengthens creative self-efficacy. Positive emotions during creative work support self-efficacy, while anxiety or frustration may diminish it.

Understanding whether and how these traditional sources of self-efficacy function in AI contexts is central to this review's investigation. This review examines how the four sources operate when creative processes involve technological collaboration, and under what conditions they continue to support the development of creative confidence.

Creative self-efficacy, creative confidence, and creative agency are closely related but analytically distinct constructs that operate at different levels of creative self-beliefs. Creative self-efficacy refers to task- and domain-specific beliefs about one's capacity to perform creatively, while creative confidence reflects broader, more generalized beliefs about one's creative identity and willingness to engage in creative activity across contexts. Creative agency captures perceived control, intentionality, and ownership over creative processes and outcomes, and helps explain how task-level efficacy experiences accumulate into broader confidence over time.

Importantly, these constructs are dynamically interrelated. Mastery experiences that strengthen creative self-efficacy can contribute to the development of creative confidence, particularly when individuals perceive themselves as the primary agents of creative outcomes. Agency, therefore, functions as a critical linking construct: when individuals experience control and authorship in creative work, efficacy beliefs are more likely to generalize into sustained confidence and creative identity. This conceptual relationship provides an essential foundation for examining how human-AI collaboration may alter creative self-beliefs, as AI systems can reshape perceptions of control, attribution, and authorship in creative activity.

Understanding whether and how each of these four sources functions differently when creative processes involve generative AI collaboration is a central question of this review, examined empirically across the studies synthesized below. Of particular interest is whether AI-mediated creative contexts preserve the conditions, especially mastery experiences with clear attribution and social persuasion from trustworthy sources, through which the four sources typically build creative self-efficacy.

1.3. Generative AI and creative self-efficacy

Generative AI technologies have the potential to enhance creative self-efficacy by lowering technical barriers and extending creative possibilities. These capabilities can democratize creative practice and provide supportive scaffolding that builds confidence, particularly for novices (Habib et al., 2024). At the same time, generative AI can disrupt how creative practitioners experience agency and ownership over their work, as some professionals report identity tensions and shifting authorship boundaries when collaborating with AI tools (Arbaiza et al., 2024; Kellerwessel & Ujhelyi, 2024; Worrall & Collins, 2024).

When generative AI contributes substantially to creative outputs, questions of attribution and authorship have been noted in philosophical and ethical discussions of AI-assisted creativity (Chatterjee, 2022). If students cannot clearly identify their own contributions to creative work, the confidence-building power of mastery may be compromised. When students compare their unassisted work to AI-generated outputs, they may experience self-doubt about their creative abilities, a risk suggested by recent work on AI-supported ideation in the classroom (Habib et al., 2024). Drawing on social comparison theory, some scholars have suggested that exposure to AI-generated exemplars may function as a comparison standard, with potentially different motivational effects depending on individuals' prior confidence or skill level (Urban et al., 2024). In AI-supported creative environments, feedback may flow through algorithmic channels alongside human input, with some studies suggesting that AI-generated suggestions can carry unintended authority effects, particularly when system transparency is limited (McGuire et al., 2024). And emotional responses arise not just from individual creative struggles but from the quality and responsiveness of human-technology interaction itself.

Early research indicates that AI's impact on creative self-efficacy is not technologically determined but depends fundamentally on how human-AI relationships are designed and implemented. Studies across multiple domains show that creative self-efficacy remains a crucial mediator between technological interventions and creative outcomes (Habib et al., 2024; Pellas, 2023; Smit et al., 2025; Tao et al., 2024; Urban et al., 2024; Wong et al., 2021). These studies suggest that the psychological impact of AI is influenced by the technology's capabilities and the perceived integrity of human agency within the creative process.

1.4. Creative agency and control in human-ai collaboration

Creative agency, understood as the sense of personal control, intentionality, and ownership over one's creative processes and outcomes, is central to how individuals enact and attribute their creative behavior (Karwowski & Beghetto, 2019). When agency is clear and unambiguous, mastery experiences can be recognized as personally authored achievements, which Bandura (1977) identified as foundational to the development of self-efficacy. Traditional creativity models position agency as an individual attribute:

creators initiate actions, make decisions, evaluate outcomes, and claim authorship over results. This clear attribution of agency supports self-efficacy by making mastery experiences unambiguous and personal. According to social cognitive theory, mastery experiences—where individuals can attribute success to their own efforts and skills—are a primary source of self-efficacy beliefs (Bandura, 1977).

Generative AI may challenge this assumption. When AI systems autonomously generate creative outputs, substantially shape creative decisions, or execute creative visions with minimal human intervention, attribution can become complex. Who deserves credit for an AI-assisted poem—the person who wrote the prompt, or the system that generated the language? What constitutes personal mastery when technical execution is machine-mediated? How do creators maintain creative ownership and identity when collaborating with systems that may surpass their unaided capabilities? These questions suggest that agency and control dynamics may critically mediate how AI affects creative self-efficacy. Whether AI enhances or undermines confidence may depend less on the technology's capabilities than on how human-AI collaboration is structured to preserve or displace human agency—a hypothesis this review investigates empirically across educational, professional, and artistic domains.

1.5. Environmental and contextual influences

Creative self-efficacy develops within organizational, educational, and cultural contexts that shape both individual beliefs and the implementation of technology. In educational settings, teachers' attitudes toward creative risk-taking, pedagogical approaches that emphasize process over product, and feedback practices that support experimentation all influence students' development of creative self-efficacy (Beghetto, 2010). As AI tools become integrated into creative learning environments, these established pedagogical factors interact with technological design choices. Educators who frame AI as a creative partner, while emphasizing human creative agency, may foster different self-efficacy outcomes than those who position AI as an autonomous creative tool or a productivity shortcut.

In professional and organizational contexts, leadership practices and organizational culture shape whether psychologically safe learning environments are established (Cha & Edmondson, 2006). These conditions may, in turn, influence how individuals interpret AI-supported work in relation to their creative roles and professional identities. Transformational leadership and supervisory support predict higher creative self-efficacy in traditional settings; whether these same factors buffer against potential erosion of AI-related confidence remains an open empirical question. Additionally, professional creative identity—the extent to which individuals define themselves through their creative expertise—may be particularly vulnerable when AI systems demonstrate capabilities that challenge domain-specific mastery. Understanding these environmental and individual factors is essential for interpreting the varied effects of AI on creative self-efficacy observed across domains. Interpretations derived from these varied effects are applied throughout this review to inform evidence-based recommendations for educational practice, where the conditions shaping creative self-beliefs, pedagogical framing, feedback design, and human agency are most directly amenable to intervention.

2. Objectives

This review covers research published between January 2021 and March 2025. Although algorithmic and generative systems capable of producing creative outputs existed prior to this period—including GAN-based image synthesis tools and early large language models—the years covered represent a materially distinct phase characterized by non-expert accessibility, mainstream adoption, and everyday exposure. Specifically, DALL-E 2 (OpenAI, April 2022), Stable Diffusion (Rombach et al., 2022), and Midjourney (July 2022) introduced high-quality image generation to broad public audiences; ChatGPT (November 2022) achieved rapid and widespread public adoption, becoming routinely integrated into educational platforms, creative software, and professional workflows within months of its release (Cao et al., 2023); and by 2023–2024, generative AI tools were embedded in everyday creative and learning environments across domains.

This period is analytically distinct because the psychological and pedagogical conditions changed fundamentally: individuals were now routinely exposed to AI-generated creative artifacts, engaging in everyday human–AI co-creation, and encountering generative AI systems in formal learning environments. These are precisely the conditions relevant to the study of creative self-beliefs. Focusing on this period, therefore, allows the review to examine CSE, CC, and CA as they are being reshaped by generative AI under the conditions in which most people now encounter these technologies.

The following research questions guide this review:

RQ1: How does generative AI influence creative self-efficacy and confidence, and what psychological mechanisms mediate these effects?

RQ2: How do these generative AI effects vary across domains, and what strategies best support human creative agency in those environments?

3. Methodology

This systematic review was conducted per Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021) to ensure transparency and rigor in identifying, screening, and synthesizing relevant literature. PRISMA is a standardized reporting framework that helps researchers conduct and document systematic reviews with greater transparency and methodological rigor. The guidelines include a detailed checklist and flow diagram that ensure key methodological steps are clearly

reported, from the search strategy through to the final study selection. Following PRISMA standards enables readers to better evaluate the quality and completeness of the systematic review methodology.

A comprehensive literature search was performed between January 2021 and March 2025 across four major academic databases: ProQuest, Scopus, Web of Science, and Google Scholar. This timeframe was selected to capture research emerging during a period of increased accessibility and mainstream use of generative AI systems across domains.

Search terms were structured into three conceptual categories:

1. **Creative self-belief constructs:** “creative confidence” OR “creative self-efficacy” OR “creative agency”
2. **AI-related terms:** “generative AI” OR “artificial intelligence” OR “AI” OR “ChatGPT”
3. **Creativity-related outcomes:** “creativity” OR “creative thinking” OR “innovation” OR “problem-solving”

Boolean operators (AND/OR) were used to combine terms across categories. The search strategy incorporated both system-level descriptors (e.g., “generative AI,” “AI-assisted creativity”) and selectively chosen platform names to improve the retrieval of studies that explicitly reference specific tools. “ChatGPT” was included as a supplementary search term based on a methodological principle of operational label coverage: systematic searches should incorporate the labels researchers actually use when reporting their studies, to ensure that empirical work using platform-specific terminology is not excluded by searches relying exclusively on category-level terms such as “generative AI” or “large language model.” During the review period, “ChatGPT” was the most standardized platform-level label in studies examining creativity, learning, and related psychological constructs — a pattern confirmed by preliminary scoping searches conducted prior to the main search. The inclusion of “ChatGPT” as a search term therefore improved retrieval sensitivity for studies that may not have used generic category terms, without restricting the scope of the review to ChatGPT-specific findings; all retrieved studies were evaluated for relevance to creative self-beliefs regardless of which AI system they examined.

Because Google Scholar does not provide structured filtering comparable to bibliographic databases, unfiltered keyword searches

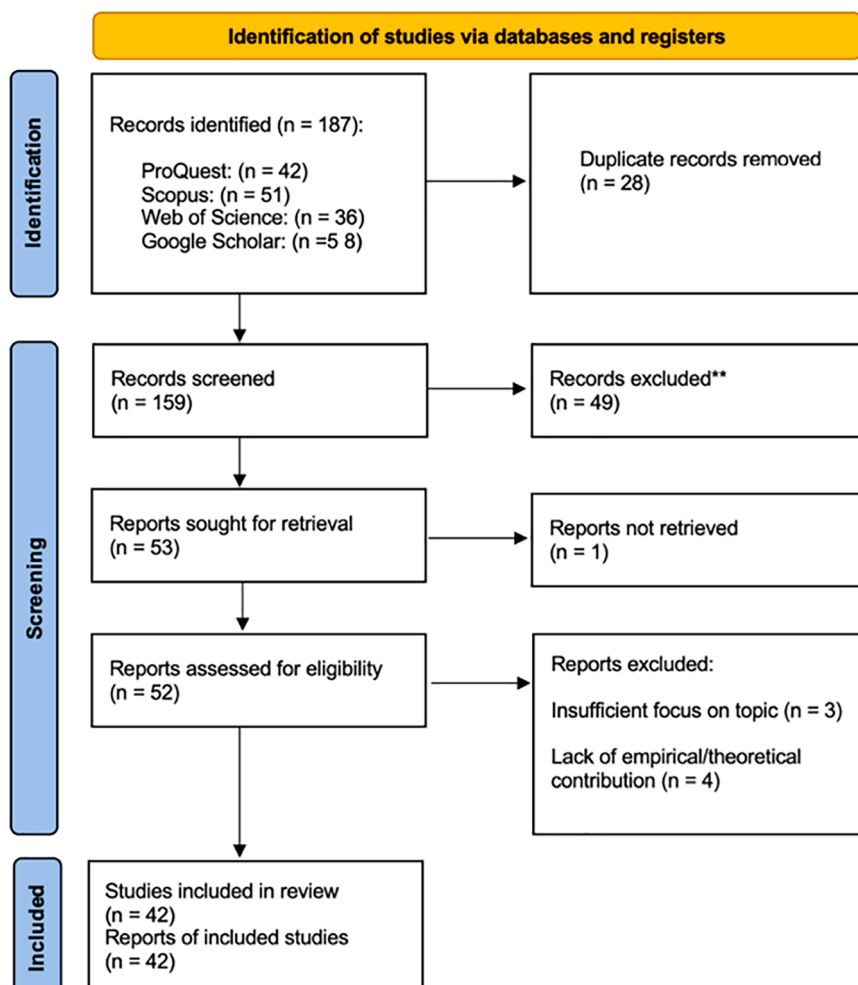


Fig. 1. PRISMA flow diagram of study selection process.

can yield several thousand results. In this review, Google Scholar was queried using the same parameters and publication date limits (2021–2025) as across the other databases. Under these constraints, the initial retrievable set from Google Scholar consisted of 58 records. These records were pooled with results from the other databases and subjected to the same deduplication and screening procedures as all other sources, as illustrated in the PRISMA flow diagram.

The ACM Digital Library is the principal indexing source for HCI research published in venues such as CHI and CSCW. The current review did not directly search the ACM Digital Library; instead, Google Scholar was used to capture interdisciplinary work, including work from HCI venues. While Google Scholar indexes a substantial proportion of ACM conference proceedings, and several CHI and CSCW papers were identified through this approach, full systematic coverage of ACM-indexed venues cannot be guaranteed through Google Scholar alone. This decision reflects the primary disciplinary framing of this review within psychology and education, for which ProQuest, Scopus, and Web of Science provide authoritative coverage. The potential implications of this decision for completeness are acknowledged in the Limitations section.

3.1. Study identification and screening

The database searches yielded a total of **187 records** (42 from ProQuest, 51 from Scopus, 36 from Web of Science, and 58 from Google Scholar). All records were imported into a reference management system, and **duplicates were removed ($n = 28$)**, resulting in **159 unique records**.

Screening followed PRISMA guidelines and occurred in two stages:

- 1. Title and abstract screening** Two independent reviewers screened all 159 records against predefined inclusion criteria. Records were excluded if they did not address creative self-efficacy, creative confidence, or creative agency, or if they did not engage with AI-related contexts. Discrepancies were resolved through discussion until consensus was reached.
- 2. Full-text review** Following title and abstract screening, **53 articles** were retained for full-text review. Full texts were independently reviewed by the same two reviewers. During full-text screening, studies were excluded for insufficient focus on creative self-beliefs, a lack of empirical data or substantive theoretical contribution, or inaccessibility of the full text. The final sample consisted of **42 studies** that met all inclusion criteria.

Fig. 1 presents the PRISMA flow diagram detailing the study selection process.

3.2. Inclusion and exclusion criteria

Studies were included if they met all of the following criteria:

Table 2
Findings for RQ1 – AI's influence on creative self-efficacy and confidence.

Author (Year)	Focus / Domain	Key Findings	Methods
McGuire et al. (2024)	Creative Professionals / General	AI-human co-creation enhances creative self-efficacy; transparency critical	Experimental; mixed methods
Newman et al. (2024)	Children (Educational)	Scaffolded AI interactions support children's creative self-efficacy through agency	Experimental intervention
Urban et al. (2024)	University Students	AI collaboration improved problem-solving and creative self-efficacy, especially in low-confidence users	Experimental
Kellerwessel and Ujhelyi (2024)	General Population	AI art creation influenced creative identity; higher self-efficacy maintains stronger identity	Qualitative
(Li et al., 2024)	Professional Digital Artists	Intervenable AI systems preserved creative self-efficacy through maintained control	Mixed methods
Tao et al. (2024)	Workplace / Various	Cognitive outsourcing correlates with creative self-efficacy; balance of delegation crucial	Scale development; quantitative
Pellas (2023)	Students (Educational)	Generative AI enhanced writing self-efficacy; effects moderated by prior digital literacy	Experimental
Kong and Yang (2025)	Secondary and University Students	AI literacy interventions improve AI empowerment and creative self-efficacy	Survey; experimental
Bueno et al. (2025)	General Population	Shift from individual to relational creativity with AI; creative self-efficacy arises from human-AI assemblages	Theoretical review
Worrall and Collins (2024)	Professional Musicians	AI music tools challenge creative identity; higher self-efficacy linked to strategic adaptation	Qualitative
Chen, E. et al. (2024)	Neurodiverse Learners	AI enhances creative self-efficacy with personalization and perceived control	Mixed methods
Arbaiza et al. (2024)	Advertising Professionals	Creative self-efficacy mediates AI adoption; ethical concerns and tech literacy affect confidence	Surveys; qualitative
Chatterjee (2022)	Artists	AI challenges traditional notions of creative agency; proposes new artist-AI relationship frameworks	Theoretical review
Lee (2022)	Creative Industries / General	AI shifts creativity discourse to sociotechnical systems; creative identity reconceptualized as distributed agency	Theoretical review

1. Focused on creative self-efficacy, creative confidence, or creative agency
2. Were peer-reviewed journal articles or peer-reviewed conference proceedings
3. Published in English between 2021 and 2025
4. Provided empirical evidence or substantive theoretical contributions relevant to understanding creative self-beliefs in contemporary contexts involving AI or during the AI integration period

Given the emerging nature of research at the intersection of AI and creativity, the included studies were categorized as either:

- Directly examining generative AI and creative self-beliefs, or
- Providing contextual insight into creative self-efficacy or confidence during the AI integration period

This dual categorization enables transparency about the limited direct evidence base while situating AI-related findings within the broader creative self-beliefs literature.

3.3. Data extraction and analysis

Data analysis followed Braun and Clarke's (2006) analysis approach. A standardized data extraction table (see [Tables 2–4](#)) was used to systematically organize publication details (authors, year, journal); study focus (creative self-efficacy, AI, intersection of both); research context (educational, organizational, arts/media); and study type (empirical, theoretical, review). The classifications provided a framework for synthesizing findings across studies to address the research questions.

3.4. Thematic analysis process

The thematic analysis was conducted following the six phases outlined by [Braun and Clarke \(2006\)](#). These steps mirror systematic review practices for transparency and replicability ([Barrett et al., 2021](#)), while allowing adaptation to the unique characteristics of AI-related creativity research. The thematic analysis proceeded through the following phases:

1. **Familiarization:** All included studies were read multiple times to gain an overall understanding of content and context.
2. **Initial coding:** Relevant text segments were coded to identify references to creative self-beliefs, agency, attribution, control, identity, and contextual variation.
3. **Theme development:** Codes were collated into preliminary themes aligned with the research questions.
4. **Theme review:** Candidate themes were reviewed against coded data and the full dataset to ensure coherence and distinctiveness.
5. **Theme definition and naming:** Themes were refined, clearly defined, and mapped onto the guiding research questions.
6. **Synthesis and reporting:** Final themes were synthesized into the Results and interpreted in the Discussion in relation to existing theory.

All coding decisions involved analytic judgment, and methodological quality varied across studies. Themes are therefore reported

Table 3
Findings for RQ2 – Contextual variation across domains (13 studies).

Authors (Year)	Focus / Domain	Key Findings	Methods
Payne and Whitworth (2022)	Higher Education (Biochemistry)	Scaffolded innovation tasks increased creative self-efficacy	Intervention study
Creely et al. (2021)	K-12 Teachers	Teacher creative self-efficacy influences risk-taking; technology enables agency	Qualitative case study
Brainard (2025)	General Population	Explores creative agency as curiosity missing in AI	Theoretical
Pireddu (2024)	General / Philosophical	Post-medial understanding of generativity and creative agency	Theoretical review
Vianna (2023)	Computational Systems	Investigates creative disobedience framework for AI	Theoretical
Li et al. (2025)	Graduate Students (Research)	Supervisor support enhances research creativity via self-efficacy	Quantitative
Radzikowski et al. (2024)	Higher Education (Laboratory)	Transdisciplinary practices enhance creative self-efficacy and reduce gender differences	Mixed methods
Smit et al. (2025)	Education (Visual Programming)	Enjoyment increased self-efficacy and reduced gender gap	Experimental
Wong et al. (2021)	Platform Crowdworkers	Digital feedback shapes creative performance via self-efficacy	Quantitative
Shah et al. (2023)	Students (General)	Soft skills and digital competence positively affect career development through creative self-efficacy	Survey
Feniger-Schaal et al. (2022)	Arts Therapies	Transition to online therapy affected creative self-efficacy; technological adaptability important	Qualitative
Macglone et al. (2022)	Community Music	Inclusive, co-creative approaches enhance professional identity and creative self-efficacy	Qualitative
Latukeyu and Pollard (2022)	Music Education	Career design interventions improved creative self-efficacy and reduced anxiety	Mixed methods

as recurring patterns observed across the reviewed literature, rather than as causal or universal claims.

By following established systematic review protocols while adapting the approach to address the unique characteristics of this emerging research area, the methodology provides a robust foundation for synthesizing current knowledge and identifying critical gaps for future investigation.

4. Results

The systematic review identified 42 peer-reviewed articles published between January 2021 and March 2025 that addressed creative self-efficacy, creative confidence, or closely related creative self-beliefs. Of these, 14 studies (33 %) directly examined the relationship between generative AI use and creative self-beliefs. An additional 13 studies examined creative self-efficacy or confidence in contexts where generative AI was present or emerging but not explicitly treated as an independent variable. The remaining 15 studies investigated creative self-efficacy in various domains during the same period and provide contextual insight into how creative self-beliefs were being conceptualized and measured, without directly addressing AI-related effects (see [Appendix A](#)).

The distribution of studies indicates that empirical research directly examining generative AI and creative self-beliefs remains limited and heterogeneous. Publication volume increased in the later years of the review period, with 17 of the 42 studies published in 2024, reflecting growing scholarly attention to creativity-related self-beliefs during a period of expanding generative AI use.

RQ1: How does generative AI influence creative self-efficacy and confidence, and what psychological mechanisms mediate these effects? Across the subset of studies that directly examined generative AI and creative self-beliefs, three recurring outcome patterns were observed.

Positive effects on creative self-efficacy and confidence were reported in 7 of the 14 studies that directly examined generative AI and creative self-beliefs ([Chen, E. et al., 2024](#); [Kong & Yang, 2025](#); [McGuire et al., 2024](#); [Newman et al., 2024](#); [Pellas, 2023](#); [Urban et al., 2024](#); [Bueno et al., 2025](#)), particularly in educational and training-oriented contexts. In these studies, generative AI was positioned as a supportive or augmentative tool, and increases in creative self-efficacy were most often reported among novice or lower-confidence users when human direction and decision-making remained central to the creative process (e.g., [Urban et al., 2024](#); [Pellas, 2023](#); [Newman et al., 2024](#); [Kong & Yang, 2025](#); [Chen, E. et al., 2024](#); [McGuire et al., 2024](#); [Bueno et al., 2025](#)).

Negative or mixed effects were reported in 5 of the 14 studies, more frequently among those focused on professional or advanced creators ([Arbaiza et al., 2024](#); [Chatterjee, 2022](#); [Kellerwessel & Ujhelyi, 2024](#); [Li et al., 2024](#); [Worrall & Collins, 2024](#)). These studies documented experiences of identity tension, role ambiguity, or reduced perceived agency, particularly in contexts where AI systems generated substantial portions of the creative output or were positioned as dominant contributors e.g., ([Arbaiza et al., 2024](#); [Chatterjee, 2022](#); [Kellerwessel & Ujhelyi, 2024](#); [Li et al., 2024](#); [Worrall & Collins, 2024](#)). Two further studies reported theoretically complex or context-dependent outcomes that did not fit cleanly into either pattern ([Lee, 2022](#); [Tao et al., 2024](#)).

Across both outcome patterns, several recurring psychological mechanisms were emphasized by authors as shaping creative self-beliefs. Most prominently, perceived control and agency were frequently discussed as central conditions under which generative AI supported or undermined creative confidence. Studies also pointed to the role of transparency in AI systems, particularly in shaping trust and interpretation of AI-generated feedback, as well as issues of identity and authorship, especially among professional creators negotiating their creative role in relation to AI.

Two studies illustrate these contrasting patterns concretely. [Newman et al. \(2024\)](#) reported that children using an intervenable AI drawing tool demonstrated measurable gains in creative self-efficacy when the system required them to make explicit directional decisions at each step, preserving their sense of authorship. By contrast, [Worrall and Collins \(2024\)](#) found that professional musicians reported diminished creative confidence when working with AI systems that autonomously generated substantial musical material, leaving performers uncertain about the boundaries of their own contribution.

4.1. Summary of RQ1 findings

Taken together, these patterns suggest that the influence of generative AI on creative self-efficacy and confidence is not uniform, but varies systematically with how creative control, attribution, and role boundaries are structured within human–AI interaction contexts. Enhancement of creative self-beliefs was most consistently associated with conditions that preserved meaningful human agency, while neutral or negative outcomes were more frequently observed when creative authority shifted toward the AI system. Detailed study characteristics are presented in [Table 2](#).

RQ2: How do these generative AI effects vary across domains, and what strategies best support human creative agency in those environments?

4.2. Contextual and design conditions associated with differential outcomes

Across the reviewed studies, the influence of generative AI on creative self-efficacy and confidence varied systematically according to contextual setting, role of the AI system, and interaction design features. Rather than uniform effects, outcomes appeared to cluster around recurring configurations of use in the reviewed studies.

Educational and training contexts were most frequently associated with positive effects on creative self-beliefs. In these studies,

generative AI was typically integrated as a scaffolded support for learning, ideation, or exploration. Tasks were often structured to emphasize process over output quality, and learners retained responsibility for decision-making and refinement. Under these conditions, AI use was associated with increased confidence, particularly among novices or students with lower initial creative self-efficacy.

In contrast, professional and expert creative contexts more often produced mixed or negative effects on creative self-beliefs. Studies conducted with designers, artists, or advanced creators described tensions related to authorship, identity, and role boundaries when AI systems generated substantial portions of creative output. In these contexts, AI was more likely to be framed as a production partner or autonomous generator, which was associated with reduced perceived agency or uncertainty about personal creative contribution.

Across domains, the role assigned to the AI system emerged as a consistently differentiating factor. When AI was positioned as an assistive tool that generated prompts, alternatives, or feedback, creative self-efficacy was more likely to be maintained or enhanced. Conversely, when AI functioned as a primary generator or evaluator of creative output, participants more frequently reported diminished confidence or ambiguous self-assessment.

Several studies also highlighted the importance of interaction design features, particularly transparency and controllability. Designs that made AI processes, limitations, or decision logic more visible were associated with greater trust and more stable creative confidence. In contrast, opaque systems or default automation increased the likelihood that AI suggestions were treated as authoritative, shaping users' confidence judgments independently of their own creative input.

Finally, task framing and assessment criteria influenced outcomes across contexts. Studies emphasizing experimentation, iteration, and exploration tended to support positive creative self-beliefs, whereas tasks emphasizing efficiency, optimization, or comparative quality evaluation were more likely to surface confidence threats or identity concerns. As one illustration of how design conditions moderated outcomes across domains, Pellas (2023) documented that writing self-efficacy improved when AI tools were framed explicitly as supports for the student's own idea development, whereas studies in which AI was positioned as a primary generator, such as those reviewed by Arbaiza et al. (2024) in advertising contexts, more often reported ambiguous or reduced creative self-efficacy.

4.3. Summary of RQ2 findings

Taken together, these findings indicate that the impact of generative AI on creative self-efficacy and confidence is strongly conditioned by contextual and design factors. Educational settings, assistive AI roles, transparent system design, and process-oriented task framing were most consistently associated with positive creative self-beliefs. In contrast, professional contexts emphasizing output dominance, opaque automation, or role substitution were more likely to produce mixed or negative effects. Detailed study characteristics are presented in Table 3.

Across domain-specific findings, creative agency — specifically the perceived control, intentionality, and ownership individuals maintained over creative processes and outputs — emerged as the critical mechanism linking contextual and design conditions to differential creative self-efficacy outcomes. When agency was preserved through transparent, assistive AI roles and process-oriented task framing, CSE was more likely to be maintained or enhanced; when agency was displaced by autonomous AI roles or opaque system design, mixed or negative CSE outcomes were more frequently reported.

5. Discussion

The Discussion is organized into four parts. First, we interpret the overall pattern of findings in relation to Bandura's four sources of self-efficacy and the mechanisms through which generative AI appears to reshape their operation in creative contexts. Second, we examine domain-specific variation across educational, professional, and artistic contexts, drawing on findings from both RQ1 and RQ2. Third, we derive implications for educational practice, where the empirical evidence base is currently strongest. Fourth, we identify the limitations of this review and priorities for future research. Throughout, we maintain the analytical distinction between creative self-efficacy (CSE), creative confidence (CC), and creative agency (CA) established in the Background section. Where the reviewed literature provides direct evidence, we specify the construct measured by the cited study. Where findings for CSE carry theoretical implications for the broader construct of creative confidence, we make this inferential step explicit rather than treating the two constructs as equivalent.

This systematic review analyzed 42 studies examining creative self-efficacy and confidence during the 2021–2025 period of rapid integration of generative AI. Of these, 14 directly investigated AI's impact on creative self-beliefs, while 13 examined contextual variation across educational, professional, and artistic domains. Across the available studies, recurring patterns were observed: whether AI enhanced or undermined creative confidence appeared to depend primarily on how human-AI collaboration preserved or displaced human agency and control.

As established in the Background, creative agency (perceived control, authorship, and intentionality) has been proposed as an important mediating construct linking task-level efficacy experiences to broader creative confidence. When individuals retain meaningful agency in creative work, mastery experiences more reliably build task-specific self-efficacy, and accumulated efficacy experiences can, over time, generalize into broader creative confidence and creative identity. Across the studies reviewed, a recurring pattern was that studies in which generative AI preserved human agency were more likely to report enhanced CSE, whereas studies in which AI displaced human authorship or obscured individual contributions were more likely to document uncertainty or the erosion of creative self-beliefs. Whether these CSE-level changes translate into durable shifts in creative confidence remains a question that longitudinal research must address.

In studies in which users maintained meaningful control over AI systems, engaged in transparent co-creation, and could clearly attribute creative contributions, researchers more often reported greater creative confidence. Conversely, in studies in which AI

displaced human authorship, operated autonomously, or obscured individual contributions, researchers more frequently reported diminished confidence (McGuire et al., 2024; Li et al., 2024; Tao et al., 2024; Newman et al., 2024; Kellerwessel & Ujhelyi, 2024). Taken together, these findings suggest AI may alter the conditions under which Bandura's four sources of self-efficacy operate in the contexts studied.

This pattern appeared across the domains represented in the available sample and did not appear to be restricted to particular tools or user groups. Within this limited body of literature, control repeatedly emerged as an important mechanism, although confirming this interpretation will require broader empirical evidence. These patterns offer preliminary implications for the application of Bandura's self-efficacy theory in AI-rich creative contexts.

For example, several studies illustrate how design choices that preserve or compromise attribution clarity are associated with differences in creative self-efficacy outcomes. In educational contexts, Newman et al. (2024) reported that children demonstrated gains in creative self-efficacy when using intervenable AI systems that required explicit creative decisions and reflection on individual contributions, conditions that align with mastery-based interpretations of Bandura's model. Similarly, Pellas (2023) documented increases in writing self-efficacy when AI tools were framed as supports for idea development rather than as autonomous generators.

In contrast, studies in professional contexts described patterns consistent with attribution erosion. Worrall and Collins (2024) reported that professional musicians experienced diminished creative confidence when AI systems generated substantial musical material without transparent links to the artists' own decisions. Arbaiza et al. (2024) similarly found that advertising professionals reported lower creative self-efficacy when AI outputs blurred authorship boundaries and limited opportunities to claim ownership of creative outcomes. Together, these cases illustrate how attribution clarity and perceived control may function as practical instantiations of self-efficacy mechanisms in human-AI creative collaboration, as interpreted through Bandura's framework.

While this review initially examined both creative confidence and creative self-efficacy as distinct constructs, the literature predominantly focused on self-efficacy rather than the broader construct of creative confidence. This reflects both methodological evolution toward task-specific instruments and the practical advantages of measuring situational beliefs in AI contexts. The findings regarding self-efficacy, however, carry implications for understanding broader creative self-beliefs, particularly given the dynamic feedback loop between task-specific successes and generalized confidence described in our theoretical framework.

Accordingly, where the Discussion uses the term "creative confidence," this refers either to: (a) a specific general-confidence measure used in the cited study, or (b) a theoretically grounded inference about broader creative identity effects that the CSE findings suggest, given the feedback loop between task-specific successes and generalized confidence described in our theoretical framework. In all other cases, we use "creative self-efficacy" to reflect what was measured.

5.1. Control as a prominent psychological mechanism

The reviewed studies suggest that control may be one important psychological mechanism shaping these outcomes, although this interpretation remains preliminary given the limited evidence base. The studies reviewed indicated that transparent, controllable, intervenable AI systems were associated with enhanced creative confidence, while opaque, autonomous systems were associated with undermined confidence (Li et al., 2024; McGuire et al., 2024; Newman et al., 2024; Smit et al., 2025; Tao et al., 2024; Wong et al., 2021). This pattern was observed across educational settings with novice students, professional contexts with domain experts, and artistic practice with creative practitioners within the reviewed sample.

The consistency among the studies reviewed suggests that control may operate as a central psychological mechanism rather than as a function of specific technological features, user characteristics, or domain requirements, though additional research is needed to confirm this interpretation across diverse contexts. In these studies, the critical variable appeared to be not merely technological capability but how human-AI interaction was designed and implemented—whether systems preserved human decision-making authority, made their operations transparent, and supported clear attribution of creative contributions.

The reviewed studies suggest that control appears to shape whether individuals can interpret creative achievements as evidence of their own capabilities, a foundation of self-efficacy development in Bandura's model. When AI systems operated transparently, and users retained meaningful decision-making authority over creative processes, studies reported that successful outcomes could still be experienced psychologically as personal achievements. The individual selected the direction, made substantive choices, evaluated options, and guided execution, even if AI handled technical implementation or generated raw material. Under these conditions, the reviewed literature suggests the psychological link between effort and outcome can remain intact.

However, when AI operated autonomously, made consequential creative decisions without user input, or produced outputs through processes users cannot understand or intervene in, studies documented that this psychological link appeared to fracture. Success could not be clearly attributed to one's own creative capability, and the mastery experience appeared to lose its efficacy-building power.

This pattern offers one possible interpretation of the paradoxical effects observed across studies: the same AI tools enhanced confidence in some implementations while eroding it in others. The difference appeared, in this sample, to lie in how human-AI collaboration was structured. This perspective offers theoretical implications: it suggests that AI does not inherently enhance or threaten creative self-efficacy. Rather, the reviewed studies indicate that AI's psychological impact may depend heavily on design choices that either preserve or compromise the attribution clarity essential to Bandura's model.

5.2. AI's effects on self-efficacy sources

Mastery experiences—direct successes in performing creative tasks—are the most powerful source of self-efficacy in Bandura's model. In human-AI collaboration, mastery becomes complex when achievements result from human-AI collaboration rather than

purely human effort. The reviewed studies suggested that attribution clarity appeared to shape whether collaborative success built confidence (McGuire et al., 2024; Li et al., 2024). When students could identify specific creative decisions they made, problems they solved, or directions they set, even if AI handled execution, successful outcomes were reported to strengthen self-efficacy. However, when AI generated substantial portions of creative output autonomously or when the relationship between user input and system output was opaque, studies documented that students struggled to claim achievements as evidence of their own capability. This created what might be termed ambiguous mastery: objectively successful creative work that appeared to fail to build confidence because individuals could not clearly attribute success to their own creative competence. The psychological mechanism underlying traditional mastery experiences, "I succeeded, therefore I am capable," requires clear attribution.

Vicarious learning traditionally involves observing similar others successfully complete creative tasks, allowing individuals to gauge their own capabilities through social comparison (Bandura, 1977). The reviewed literature suggests AI introduces a fundamentally new category of vicarious experience: exposure to AI-generated outputs and demonstrations. This shift appeared to create complex comparison dynamics where AI capabilities can either inspire or intimidate (Kellerwessel & Ujhelyi, 2024; Urban et al., 2024). Studies indicated that the psychological impact depended heavily on framing. When AI outputs were positioned as collaborative tools that expand human creative possibilities, similar to how advanced software enables new forms of artistic expression, vicarious exposure appeared to support confidence by revealing achievable creative directions.

However, when AI demonstrations suggested human creative inadequacy or implied replacement, studies reported that they eroded self-efficacy by establishing upward comparisons with seemingly superior creative agents. Notably, the comparison mechanism appeared to differ from traditional peer observation because AI capabilities are not bound by human constraints. This makes the "similarity" component of vicarious learning, central to its effectiveness in Bandura's model, problematic. Individuals may struggle to assess their own creative potential when comparing themselves to systems operating on fundamentally different principles.

Social persuasion encompasses feedback, encouragement, and evaluative messages that influence beliefs about creative potential. In human-AI collaboration, the reviewed studies indicate that persuasion increasingly flows through algorithmic channels as AI systems provide suggestions, critiques, and guidance throughout creative processes (Wong et al., 2021; Newman et al., 2024). Unlike human feedback, studies suggest AI input may be perceived as neutral, objective, or authoritative, potentially amplifying its psychological influence on creative confidence. This perceived objectivity appeared to create both opportunities and risks in the contexts studied. The authority attributed to AI persuasion appeared to depend on user AI literacy and domain knowledge. Novices may accept AI feedback uncritically, allowing algorithmic persuasion to shape self-beliefs. Experts with well-developed domain knowledge can evaluate AI suggestions more critically, maintaining appropriate skepticism about algorithmically generated persuasive arguments based on patterns observed in the reviewed literature.

Affective states, which are emotions, moods, and physical sensations experienced during creative work, inform judgments about creative competence in Bandura's model. The limited evidence on human-AI collaboration in the reviewed studies introduces both traditional creative emotions and novel technology-specific affective experiences. Studies reported that users report exhilaration when AI accelerates creative processes and enables new possibilities, frustration when systems behave unpredictably or require extensive prompt engineering, anxiety when outputs blur boundaries between human and machine contributions, and satisfaction or disappointment based on the responsiveness and quality of AI collaboration (Smit et al., 2025; Creely et al., 2021).

These emotional responses appeared to be shaped by multiple factors, including system design quality, user expectations, prior experience with AI tools, and individual differences in technology attitudes. The reviewed studies suggest that affective states in AI-mediated creativity emerge not solely from individual psychological processes but also relationally, through the dynamic between human intentions and AI responses. Smooth, responsive collaboration that maintains user agency has been reported to have a positive effect on self-efficacy. Frustrating, opaque, or unpredictable AI behavior appeared to generate negative affect that may diminish creative confidence even when technical outcomes were objectively successful.

These findings from the reviewed studies suggest that Bandura's four sources remain relevant in AI contexts but may operate differently than in purely human creative processes. The transformation is not uniform across sources in the available evidence. Mastery experiences and attribution clarity emerged as most sensitive to AI design choices; small changes in how systems present contributions or enable intervention meaningfully affected whether success built confidence in the contexts studied. When collaboration preserved human agency, all four sources could support the development of self-efficacy. This suggests that AI contexts may require attention to design factors that preserve the psychological mechanisms: clear attribution, authentic agency, and meaningful choice through which the traditional four sources build confidence based on the patterns observed in this review.

5.3. Domain-Specific patterns and contextual variation

In educational settings, the reviewed studies indicated that structured scaffolding and explicit control mechanisms helped students maintain agency when using AI tools, thereby enhancing creative self-efficacy (Newman et al., 2024; Pellas, 2023). The effectiveness of these approaches appeared to reflect developmental considerations: younger learners and those new to creative domains appeared to benefit from external supports that make agency explicit and visible. Studies demonstrated how educational interventions can address pre-existing disparities, with particular promise for reducing gender gaps in STEM domains where creative confidence differences often emerge early (Wingard et al., 2022; Smit et al., 2025).

Teacher attitudes toward creative risk-taking in technology-enhanced environments also shaped outcomes (Creely et al., 2021), suggesting that educators' own creative self-efficacy and comfort with AI tools influence how they implement these technologies with students. Educational implementations that framed AI as a creative partner while explicitly preserving students' sense of creative ownership tended to be associated with more positive creative self-beliefs in the reviewed studies. This aligns with developmental

scaffolding principles where external support gradually transfers agency to learners rather than substituting for it.

In professional contexts, studies documented that identity preservation and psychological safety emerged as central concerns. Creative practitioners often faced difficulty negotiating agency as generative AI challenged established professional identities and expertise (Worrall & Collins, 2024). The threat to self-efficacy appeared most acute when AI capabilities directly competed with skills that define professional creative competence. For instance, graphic designers are facing AI that generates publication-ready layouts, or copywriters are encountering systems that produce polished marketing copy.

Organizational factors appeared to impact these effects in the reviewed literature. Companies that adopted clear ethical frameworks for AI use, built employee AI literacy, and promoted leadership practices reinforcing creative confidence during technological transitions were reported to help protect professional self-efficacy (Dar et al., 2022; Kong & Yang, 2025; Lim et al., 2024). The findings across the reviewed studies suggest that professional creative self-efficacy in AI contexts may depend not only on individual psychological factors but also on how organizations implement technology, whether AI is positioned as threatening job security and expertise or as augmenting professional capabilities while preserving human creative authority.

Artistic domains showed complex dynamics involving deeply personal creative identities (Feniger-Schaal et al., 2022; MacGlone et al., 2022). For artists, creativity is often identity-defining rather than instrumental. Studies reported that AI collaboration raised questions about authenticity, creative voice, and what constitutes "real" artistic practice. Some practitioners experienced AI tools as liberating, expanding creative possibilities and reducing technical barriers. Others felt AI threatened the essence of artistic identity, the sense that creative work expresses unique human experience and vision. These divergent responses likely reflect individual differences in creative identity centrality and beliefs about the nature of creativity. Those with growth mindsets and pragmatic orientations toward creative tools appeared to adapt more readily (Dweck, 2006). This domain variation observed across studies suggests that AI's impact on creative self-efficacy cannot be understood through universal principles alone but may require attention to how creativity functions psychologically and socially within specific professional and artistic communities.

5.4. Implications for educational practice

Although the reviewed studies span domains, the implications discussed below focus specifically on educational practice, where the empirical evidence base is currently strongest. For educators and curriculum designers concerned with fostering creative thinking, these findings suggest that preserving agency may be an important design consideration. When integrating AI tools into creative learning activities, pedagogical designs should prioritize maintaining student control and distinguishing between human and AI contributions. This might involve structuring assignments in which students explicitly identify which creative elements they generated and which came from AI assistance, thereby preserving the attribution clarity necessary for mastery experiences to build self-efficacy.

The reviewed literature suggests that how AI is framed may shape creative self-efficacy outcomes (Pellas, 2023; Newman et al., 2024; McGuire et al., 2024; Urban et al., 2024). Positioning AI as a supportive partner in creative exploration, rather than as an autonomous creator or a replacement for human creativity, may shape students' relationships with these tools. Using language like "working with AI" rather than "using AI" or "AI-generated" emphasizes collaborative agency. Explicitly discussing what AI can and cannot do, and why human creativity remains essential, may help students maintain realistic and positive self-beliefs about their own creativity. Exercises in critically analyzing AI outputs will help establish and sustain that practice. Clear policies on attribution and plagiarism directly address authorship concerns that would otherwise undermine confidence, particularly when originality is linked to evaluation.

Effective AI integration in creative education may require progressive scaffolding that explicitly supports students' developing sense of agency in directing AI systems. Younger learners and those new to creative domains appeared to benefit from highly guided experiences where they make deliberate choices about when and how to use AI tools, with control gradually released as confidence develops (Newman et al., 2024). Beginning with "AI as research assistant" before progressing to "AI as creative collaborator" may allow students to build confidence incrementally. Concurrent development of critical evaluation skills appears essential based on the reviewed studies. Students' prior digital literacy and understanding of AI capabilities appeared to moderate effects on creative self-efficacy (Kong & Yang, 2025; Pellas, 2023).

Providing explicit instruction in how AI systems work, their capabilities and limitations, and the nature of machine learning may help students maintain realistic expectations and recognize the boundaries of AI competence. Practice exercises where students identify strengths and weaknesses in AI-generated content may build confidence in their own judgment and creative capabilities, developing the critical stance necessary for productive AI collaboration.

Attention to students' emotional experiences during human-AI collaborative creative work may support the development of self-efficacy. Studies have reported that positive affect during collaboration reinforces confidence, whereas frustration or anxiety signals the need for additional support or an adjusted task design (Smit et al., 2025). Creating opportunities for students to reflect on and discuss their emotional experiences working with AI normalizes the uncertainty many feel about collaborating with AI. Helping students understand that struggling with AI tools or feeling uncertain might be a normal and potentially productive learning experience, rather than a sign of inadequacy, frames these challenges constructively.

This connects to broader pedagogical approaches emphasizing process over product in creative learning. When assessment focuses on students' creative processes, decision-making, and learning rather than only on polished final outputs, it maintains focus on mastery experiences and reduces anxiety about AI-human output comparisons. Regular formative feedback that emphasizes students' growth and unique contributions, and explicitly acknowledges what students themselves contributed to AI-assisted projects, reinforces creative self-efficacy by preserving the link between effort and achievement.

Implementing these principles requires attention to resource disparities and educator preparation. Access to AI tools and the quality

of implementation vary significantly across educational contexts. Ensuring equitable access to high-quality AI tools and providing appropriate support for all students becomes essential to prevent AI from exacerbating existing educational inequalities (Habib et al., 2024). Evidence suggests AI can benefit neurodiverse learners when designed with personalization and user control in mind (Chen, E. et al., 2024), but realizing this potential requires deliberate attention to accessibility and inclusive design. Notably, teachers' self-efficacy in creativity and comfort with AI tools influence student outcomes (Creely et al., 2021). Professional development should help educators develop their own AI literacy and creative confidence, learn strategies for scaffolding AI use in creative instruction, and understand the psychological dynamics of human-AI creative collaboration. Without educator preparation, even well-designed AI tools may be implemented in ways that undermine rather than support student creative self-efficacy. Insights from professional and artistic contexts, particularly regarding attribution clarity, identity preservation, and perceived control, inform these educational implications by illustrating conditions under which creative self-efficacy is supported or undermined across domains.

6. Future research and limitations

This review identifies significant gaps and opportunities for advancing understanding of creative self-efficacy in the AI era. Given the limited empirical base, establishing research priorities is essential. Most current studies are cross-sectional, underscoring the need for longitudinal designs that examine how creative confidence develops through sustained engagement with AI tools.

Future work should develop and validate instruments tailored to human-AI collaboration. Expanding research to more diverse populations is another critical priority. Current studies focus largely on higher education and professional settings, with little attention to children, older adults, or neurodivergent individuals. Including these groups would provide a more comprehensive understanding of how AI influences creative efficacy across life stages and abilities. Cross-cultural research is also needed to reveal whether AI-mediated self-efficacy manifests differently across sociotechnical environments.

This review has several limitations. Our synthesis was based on 42 articles, of which only 14 directly investigated the impact of generative AI on creative self-efficacy. The heterogeneity in methods, populations, contexts, and AI systems studied limits the strength of conclusions that can be drawn. Patterns identified should be understood as emerging from this specific sample rather than as established universal principles. The predominance of educational and organizational contexts may limit generalizability to artistic practice or everyday creativity. The rapid evolution of AI technologies also means that research quickly becomes outdated; many earlier studies do not account for the latest generative systems.

A further limitation concerns database coverage. The decision to rely on Google Scholar rather than the ACM Digital Library as the source for HCI-venue studies means that full systematic coverage of CHI, CSCW, and related ACM proceedings cannot be confirmed. Future systematic reviews at this intersection of psychology, education, and human-computer interaction should include the ACM Digital Library as a primary search source to strengthen coverage of interaction design and HCI-oriented creativity research.

Finally, future reviews should incorporate non-English publications to broaden cultural perspectives. Despite these limitations, this review offers a timely synthesis of emerging evidence, identifying patterns and gaps to guide future work on how human creative self-efficacy evolves in increasingly AI-mediated environments.

7. Conclusion

This systematic review of 42 studies published between 2021 and 2025 describes an emerging research landscape examining how AI may influence creative self-efficacy and confidence. Findings from the reviewed studies suggest that AI's impact is complex and context-dependent, with both enhancement and erosion effects observed across educational, professional, and artistic domains represented in this literature.

Control and agency repeatedly emerged as prominent factors across the reviewed literature: creative self-efficacy appeared to strengthen when human-AI collaboration preserved meaningful control, maintained clear attribution, and supported creative identity. Conversely, confidence appeared more likely to weaken when AI displaced human authorship, operated autonomously, or obscured individual contributions.

For educators and curriculum designers, these findings provide both encouragement and caution. AI tools offer genuine potential to scaffold creative learning, reduce technical barriers, and democratize access to creative practice. However, realizing this potential may require deliberate attention to preserving student agency, framing AI appropriately, and supporting mastery experiences central to the development of creative self-efficacy. Given the emerging and heterogeneous nature of the evidence base, these conclusions should be understood as provisional patterns requiring further empirical validation across diverse contexts and populations.

AI statement

During the preparation of this work, the authors used Zotero for citation management and Grammarly (non-generative) for spelling and grammar checking. In preparing the second revision (Round 2) of this manuscript, an AI writing assistant (Claude.ai, Anthropic) was used to assist with paraphrasing, sentence-level rewording, and organizational restructuring of selected passages, including portions of the Results and Discussion sections. AI outputs were reviewed, substantially revised, and evaluated by the authors, who take full responsibility for all intellectual content, interpretations, citations, and conclusions presented. AI tools were not used to generate, fabricate, summarize, or retrieve source materials; all citations reflect sources directly reviewed by the authors. The use of AI assistance in this manuscript is consistent with journal policy and is disclosed here in the spirit of transparency appropriate to a manuscript examining human-AI creative collaboration.

CRedit authorship contribution statement

Sabrina Habib: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zachary Gardiner:** Writing – review & editing, Formal analysis, Data curation.

Appendix A. Studies of Creative Self-Efficacy During the AI Integration Period (2021–2025)

The following studies examined creative self-efficacy in various contexts during 2021–2025 but did not directly address AI-creativity relationships. They provide an important baseline understanding of how creative self-efficacy research evolved during the period of rapid AI integration.

Table 4

Table 4

Studies of creative self-efficacy during the AI integration period (2021–2025).

Authors (Year)	Focus / Population	Key Findings	Methods
Álvarez-Huerta et al. (2021)	Undergraduates	Creative self-concept is positively associated with student engagement. Differences in the relationships found by gender and academic year.	Quantitative survey
Puente-Díaz and Cavazos-Arroyo (2022)	General (Creative Processes)	Metacognitive feelings influence creative self-efficacy during idea generation and evaluation	Quantitative survey
Han et al. (2022)	Graduate Students (Research)	Supervisor support boosts innovative behavior via creative self-efficacy mediation	Quantitative
Lim et al. (2024)	Organizational Leaders	Transformational leadership influences innovative behavior through creative self-efficacy	Quantitative
Dar et al. (2022)	Organizational Employees	Overqualification positively affects innovative behavior mediated by creative self-confidence	Quantitative survey
Chen, Y.-W (2024)	Entrepreneurs in Culture and Creative Parks	Creative self-efficacy predicts entrepreneurial bricolage	Quantitative
Silvia et al. (2021)	General Population	Developed Humor Efficacy Scale linked to creative self-efficacy	Scale development; Quantitative
Yang et al. (2021)	Hospitality Industry Employees	Group diversity related to service innovative behavior through creative self-efficacy	Multilevel quantitative
Liu et al. (2021)	Nursing Students	Integrated course improved critical thinking and creative self-efficacy	Mixed methods
Wingard et al. (2022)	Female STEM Students	Design thinking intervention raised creative self-efficacy	Experimental intervention Quantitative
Vinarski-Peretz and Kidron (2024)	Public Managers	Climate for innovation predicts collective implementation efficacy mediated by creative self-efficacy	Quantitative
Mahdzir and Ghani (2022)	Organizational Employees	Positive leadership influences innovative behavior via creative self-efficacy	Thematic review
Goel et al. (2024)	Remote Workers	Remote work affects well-being and creative self-efficacy	Bibliometric analysis
Ibrahim et al. (2024)	General Population	Growing importance of creative thinking and creative self-efficacy	Bibliometric analysis
Luo and Ding (2024)	Organizational Teams	Boundary spanning behavior enhances innovation via creative self-efficacy	Multilevel quantitative

Data availability

Data will be made available on request.

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